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Fabrication of diamond drum resonators via laser engraving and Si wet etching for high precision temperature sensing

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Highlights

- Nanodiamonds treated by cerium ammonium nitrate have positive charges.
- Diamond drum is fabricated by laser engraving and Si wet etching.
- Diamond drum exhibit high frequency temperature coefficient (1396ppm/K).

Abstract

Diamond has attracted considerable attention in microelectromechanical systems (MEMS) due to its outstanding mechanical, thermal, and optical properties. We report here a novel strategy for facile fabrication of diamond drum resonators on silicon (Si) wafer for high precision temperature sensing. To accomplish this, nanodiamonds (NDs) colloids were first treated with cerium ammonium nitrate, a potent single-electron oxidant, to achieve a positive Zeta potential (+37 mV). The NDs seeds were then electrostatically adsorbed onto Si (100) wafer, followed by microwave plasma chemical vapor deposition (MPCVD) at 1103 K for 2 h. A continuous, pinhole-free diamond film with uniformly distributed residual stress (~ 451 MPa) and low surface roughness (~ 70 nm) was obtained. Diamond drum resonator was fabricated by a combination of laser engraving and wet chemical etching. Circular pattern of various diameters was first engraved onto the backside of the Si wafer using a 355 nm nanosecond pulse laser. Subsequent Si wet etching with 25 wt% KOH yielded square diamond drums of various sizes. We find that the etching process reduced the residual stress by 208 MPa, and enhance the SiV photoluminescence emission by 47%. The drum resonators exhibit relatively high quality factors up to ~ 2496 and high Young's modulus of ~ 812 GPa, with a power sensitivity of 427 Hz/mW and a temperature sensitivity of 907 Hz/K (1396 ppm/K). This approach offers a rapid, cost-effective method for batch fabrication of diamond drum resonators on Si wafers.

Graphical abstract

CRediT authorship contribution statement

Declaration of competing interest

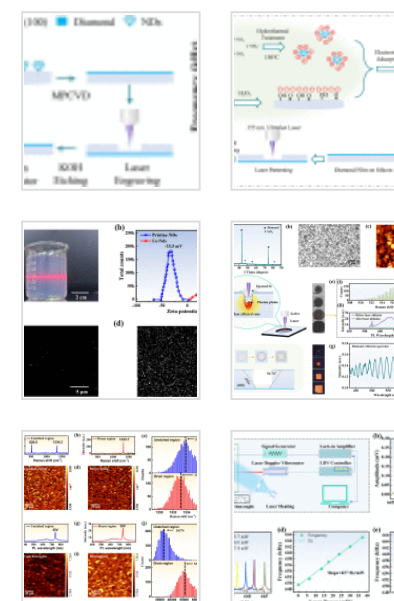
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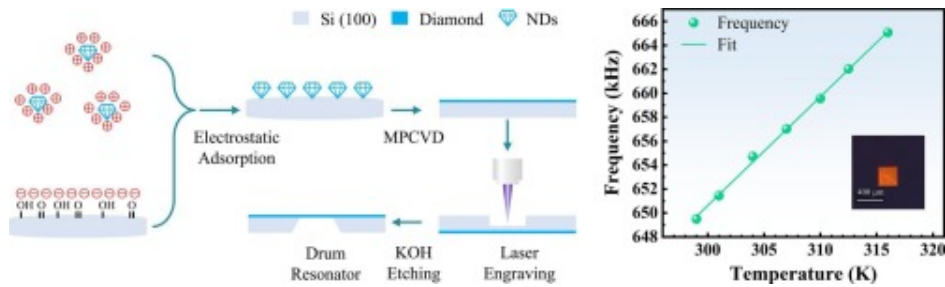
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Keywords

Diamond drum resonator; Laser engraving; Wet chemical etching; Temperature sensing

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1. Introduction

Microelectromechanical systems (MEMS) offer revolutionary potentials in a variety of fields, including sensors [1,2], actuators [3,4], biomedical devices [5,6], and telecommunications [7,8]. The choice of materials is critical to MEMS device performance. Diamond has recently attracted considerable attentions in development of miniature MEMS devices [9]. Compared to traditional MEMS materials such as silicon (Si) or silicon nitride, diamond has an exceptionally high Young's modulus ($\sim 1200\text{ GPa}$) [10] and thus support vibration at higher frequencies. Its low coefficient of friction and excellent wear resistance [11] enables stable operation of dynamic MEMS units without wear induced degradation. Besides, diamond has a high thermal conductivity ($2000\text{ W m}^{-1}\text{ K}^{-1}$) [12,13] and is of excellent mechanical and chemical inertia against degradation [14], and thus

supports stable and proper device functions under high temperature, high pressure and corrosive environments.

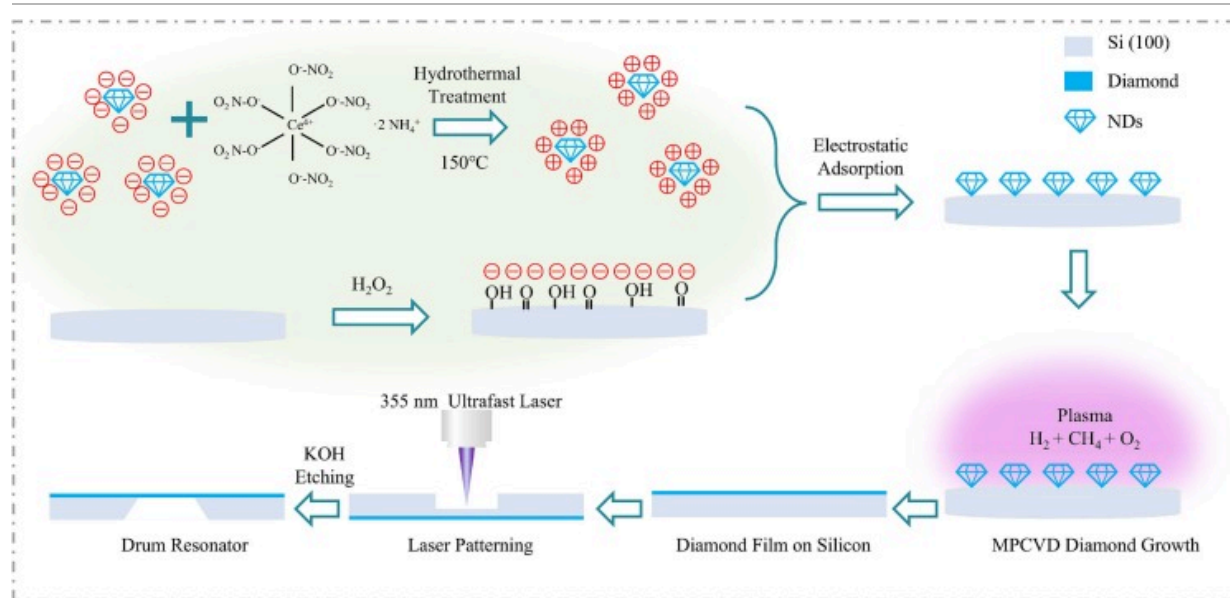
At present, fabrication of diamond MEMS devices usually involves complex procedures and sophisticated apparatus. For example, single crystal diamond MEMS mechanical resonators have been fabricated by using intelligent cutting through ion implantation assisted lifting (IAL) [15] or reactive ion thinning [16], followed by photolithography and reactive ion etching [17,18]. The process resemble those in the Si industry, but is currently restricted by the commercial availability of single-crystal diamond substrates for batch production. In contrast, polycrystalline diamond offers the advantage of large-area availability and scalability. Polycrystalline diamond MEMS resonators have been fabricated from diamond thin film by using photolithography and reactive ion etching [19], or by direct growth of diamond pattern seeded by inkjet printing [20] or micro-contact printing [21].

Laser microfabrication is a versatile technique with many advantages, such as no mask requirement, strong directivity, low cost and short processing time. It has been utilized to create intricate microscale patterns or features through laser beam manipulation on various materials such as metals [22], semiconductors [23,24], glass [25] and polymers [26]. For example, Wu et al. used femtosecond (fs) laser to obtain black Si wafer that contains a quasi-ordered array of conical microstructures, increasing its light absorption to ~90% from ultraviolet to near infrared region [23]. Laser microfabrication of MEMS resonators remains to be an area with tremendous potentials to be explored.

We present in this paper a cost-efficient and convenient method for fabricating diamond drum resonators on Si (100) wafer. The scheme involves electrostatic seeding, microwave plasma chemical vapor deposition (MPCVD) growth of smooth diamond film, laser engraving of the backside substrate, and anisotropic Si wet etching. Experimental and theoretical analyses are conducted to explore the potential applications of the diamond drum resonators in high precision temperature sensing.

2. Experimental section

Fig. 1 shows the fabrication processes of the diamond drum resonators. Firstly, the Si wafer (500 μm in thickness) was ultrasonicated in sequence in 30wt% hydrogen peroxide (H_2O_2), acetone, ethanol and deionized water, each for 5 min, respectively. Detonation nanodiamonds (NDs) particles (Zhongnan, China) were then added into a 0.05M aqueous solution of cerium ammonium nitrate ($(\text{NH}_4)_2\text{Ce}(\text{NO}_3)_6$, Aladdin, China) and ultrasonicated for 30min with optimized pH value. The mixture was then hydrothermally treated in a steel autoclave at 150 $^\circ\text{C}$ for 2h to obtain stable Ce-NDs colloid solution. The colloid was ultrasonicated for another 30min. We then measured the Zeta potential and hydrodynamic radii of the NDs colloids by dynamic light scattering (Zetasizer Advance, Malvern, USA). Electrostatic seeding was realized by immersing the Si (100) wafer into the Ce-NDs colloid. The samples were then rinsed with deionized water and dried using gentle N_2 blow. Our experiments revealed that electrostatic seeding for several seconds is usually sufficient to ensure growth of continuous diamond film.



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Fig. 1. Fabrication of diamond drum resonators through sequential procedures that include electrostatic seeding, MPCVD growth, laser engraving, and chemical etching release.

The NDs seeded Si wafer was then laser cut to obtain substrates of $5 \times 5 \text{ mm}^2$ in sizes for diamond growth. The diamond growth was realized in an MPCVD apparatus (Uniplasma, UP-206, China). The growth process had been optimized previously and lasted 2 h. During growth, the substrate temperature was set to be 830°C , the flow rates of H_2 , CH_4 , and O_2 was set to be 500, 20, and 0.5 sccm, respectively. The chamber pressure and the microwave power were maintained at 100 Torr, and 3300 W, respectively.

The crystallinity of the diamond films was evaluated by using an X-ray diffraction spectrometer (XRD. D8 Advance X, Bruker, USA). The surface morphology was characterized by scanning electron microscopy (SEM, Hitachi S-4800, Japan). The residual stresses, photoluminescence (PL), optical reflectivity, and surface roughness of the samples were evaluated by using confocal Raman spectroscopy (alpha-300RS, Witec, Germany) equipped with an atomic force microscopy (AFM) unit.

To fabricate the diamond drum resonator, the backside of the Si substrate was selectively engraved by using a 355 nm ultraviolet (UV) nanosecond laser with a repetition rate of 50 kHz and a single pulse energy of 0.248 mJ. This process created circular holes of ~ 400 , 600, 800, 1000 μm in diameters and a depth of $\sim 400 \mu\text{m}$. The samples were then subjected to etching at 80°C for 1.5 h in 25 wt% KOH aqueous solution mixed with 10 vol% isopropyl alcohol.

The dynamic frequency responses of the diamond drum resonators were measured by a Doppler vibrometer (OFV-5000/534, Polytec, Germany) equipped with a high-precision, low-noise lock-in amplifier (MFLI5M, Zurich Instruments, Switzerland). The temperature dependent responses of the resonators to optical irradiation were evaluated by directing a 450 nm continuous laser beam on the sample surface. A power meter (ThorLabs PM100D, USA) was used to calibrate the laser power, and a micro thermocouple (Aidiwen HH-K-24,

China) was employed to measure the sample temperatures. The characteristic frequency analysis of diamond drum is carried out by finite element calculations using the solid mechanics module of the commercial COMSOL multi-physics simulation software.

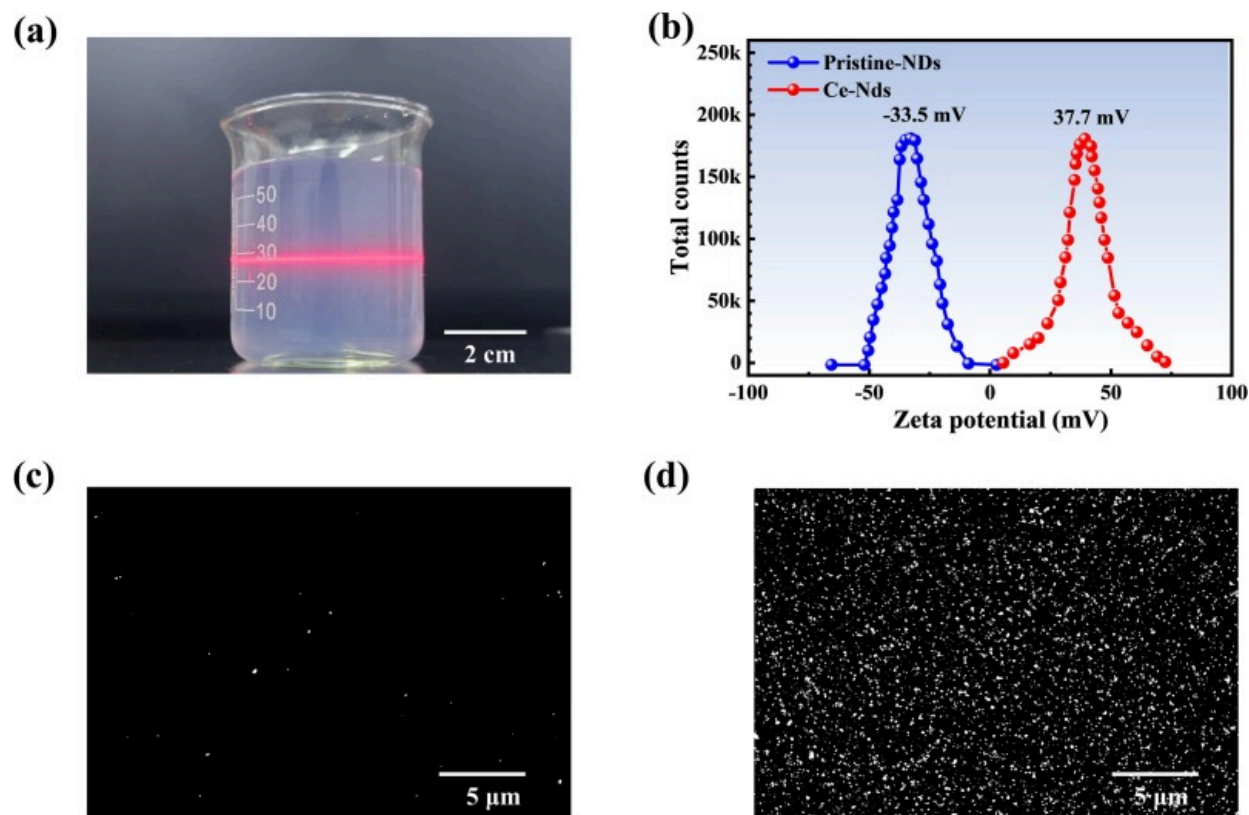
3. Results and discussion

3.1. Electrostatic seeding

Diamond nucleation on heterogeneous substrates is energetically unfavorable [27]. Electrostatic seeding is a generic technique to facilitate growth of smooth diamond thin film [28]. To achieve high density and even electrostatic seeding, NDs particles should be dispersed in solutions to form stable colloid. As detonation NDs particles are covered by carboxyl groups [29], they are usually negatively charged and cannot be directly seeded onto negatively charged substrates. Various attempts have been exerted to address this issue. For instance, Girard et al. [30] achieved a seeding density of $3 \times 10^{11} \text{ cm}^{-1}$ by electrostatic adsorption of NDs onto substrates coated with cationic polymers. Williams et al. obtained highly stable, monodisperse 3–4 nm NDs colloids with Zeta potentials exceeding +60 mV [31] and achieved a seeding density as high as 10^{12} cm^{-1} [32]. In this work, we modified the NDs particles to become positively charged by coordination with high-valence transition metal cations [33].

The Deryaguin-Landau-Verwey-Overbeek (DLVO) theory [34,35] indicates that the interaction between colloidal particles is dominated by electrostatic repulsion and van der Waals attraction. The balance of the two forces and the stability of the colloid are determined by the Zeta potential at the solid/liquid interface. We achieved well-dispersed Ce-NDs colloid by ultrasonication of NDs particles in $(\text{NH}_4)_2\text{Ce}(\text{NO}_3)_6$ solution at pH=4 for 0.5 h, followed by hydrothermal treatment. After treatment, we observed a clear Tyndall effect (Fig. 2a) by shining a 633 nm laser through the colloid solution. We then measured the Zeta potential of the solution by dynamic light scattering. Fig. 2b shows that the Zeta potential was tuned from initial -33.5 mV for the raw NDs in water to +37 mV. After treatment, the average NDs particle size was calculated to be ~110 nm by using the

Stokes-Einstein equation [36]. Although Si (100) wafer is naturally negatively charged in water, it cannot guarantee stable and uniform electrostatic seeding. In our experiments, the Si (100) substrates were also ultrasonicated in H_2O_2 solution. This process ensures formation of a thin SiO_2 layer with an isoelectric point of ~ 2.9 , which forms a negatively charged surface at $\text{pH}=4$ in water [37,38].



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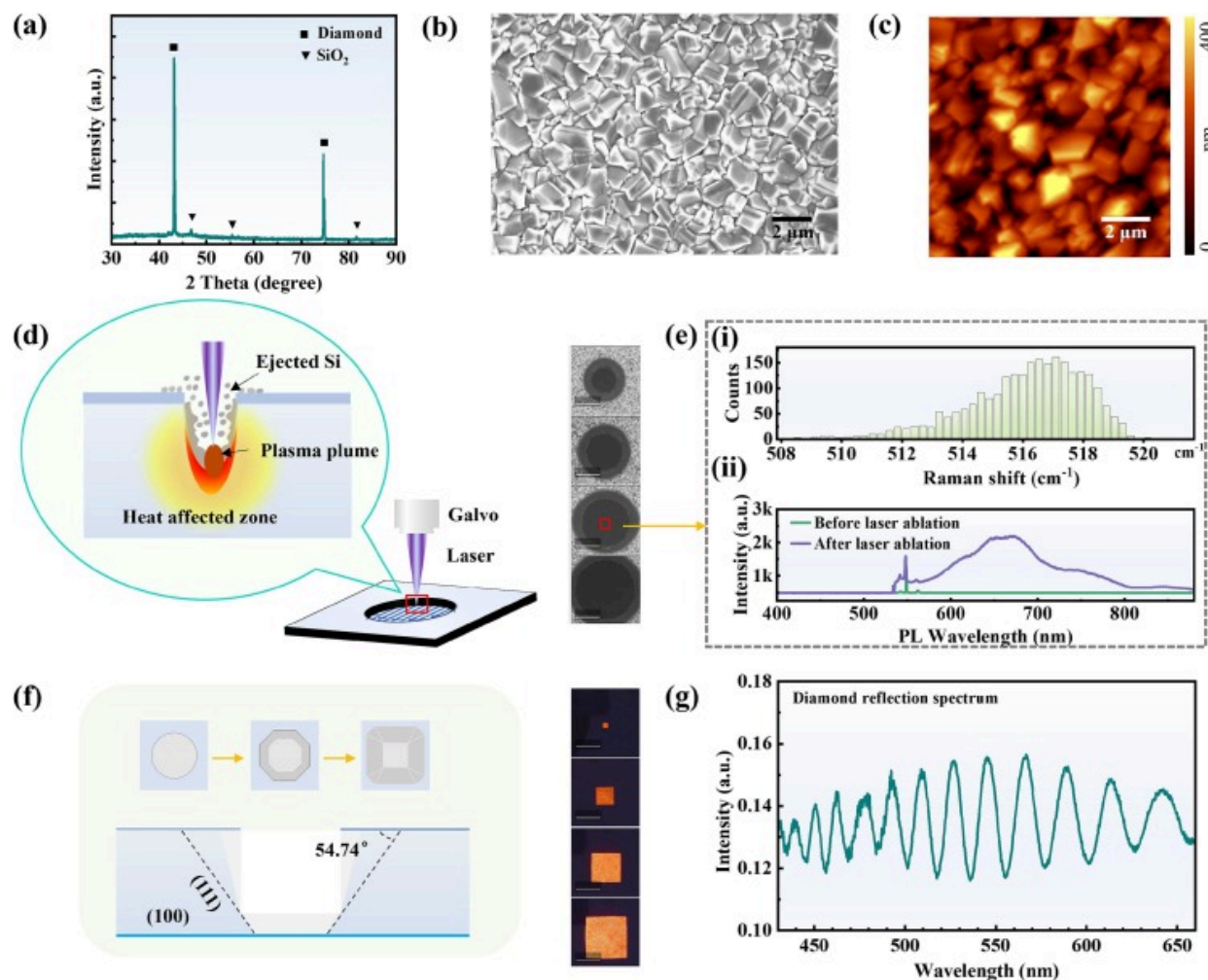
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Fig. 2. (a) The Ce-NDs colloid with Tyndall effect. (b) Zeta potentials of the pristine NDs colloid and the Ce-NDs colloid. SEM images reveal distinct seeding densities on Si substrate by NDs (c) and Ce-NDs (d) colloids.

Fig. 2c and d show the SEM images of the sample surfaces after seeding with NDs and Ce-NDs colloids. Obviously, the usage of Ce-NDs colloid results in a higher density seeding than the NDs colloid. Assuming that NDs particles are spherical, the maximum theoretical seeding density n is calculated by the formula $n = 2/\sqrt{3}d^2$ [39], where d is the particle diameter. For $d = 110\text{nm}$, the maximum seeding density is calculated to be $\sim 10^{10}\text{cm}^{-2}$. A statistical analysis shows that the seeding density in Fig. 2d approaches $\sim 10^9\text{cm}^{-2}$, closely matching the theoretical value.

3.2. Diamond film characterization

Fig. 3a shows the XRD pattern of the as-grown diamond film, which reveals a (111) diffraction peak at 43.3° and a (220) diffraction peak at 74.8° . In addition, weak diffraction peaks of silica at 46.7° , 55.3° and 81.0° were also observed. The SEM image shows the diamond film is continuous and pin-hole free with a grain size of $1\mu\text{m}$ (Fig. 3b). The AFM image shows a root-mean-square (RMS) roughness of $\sim 70\text{nm}$ (Fig. 3c).



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Fig. 3. (a) XRD pattern of the as-grown diamond film. (b) SEM image of the diamond film. (c) AFM image of the diamond film over a $10 \times 10 \mu\text{m}^2$ area. (d) Schematic illustration of laser engraving, and optical images of four holes of different diameters formed by laser engraving on Si. The scale bar is $400 \mu\text{m}$. (e) Statistic distribution of the Raman peak positions of the laser engraving region (i) and the PL spectra before and after laser ablation (ii). (f) Schematic illustration of anisotropic etching of (100) Si substrate by KOH

and optical images of square shaped diamond drums obtained after KOH etching. The scale bar is 400 μm . (g) A typical optical reflection spectrum obtained on diamond drum.

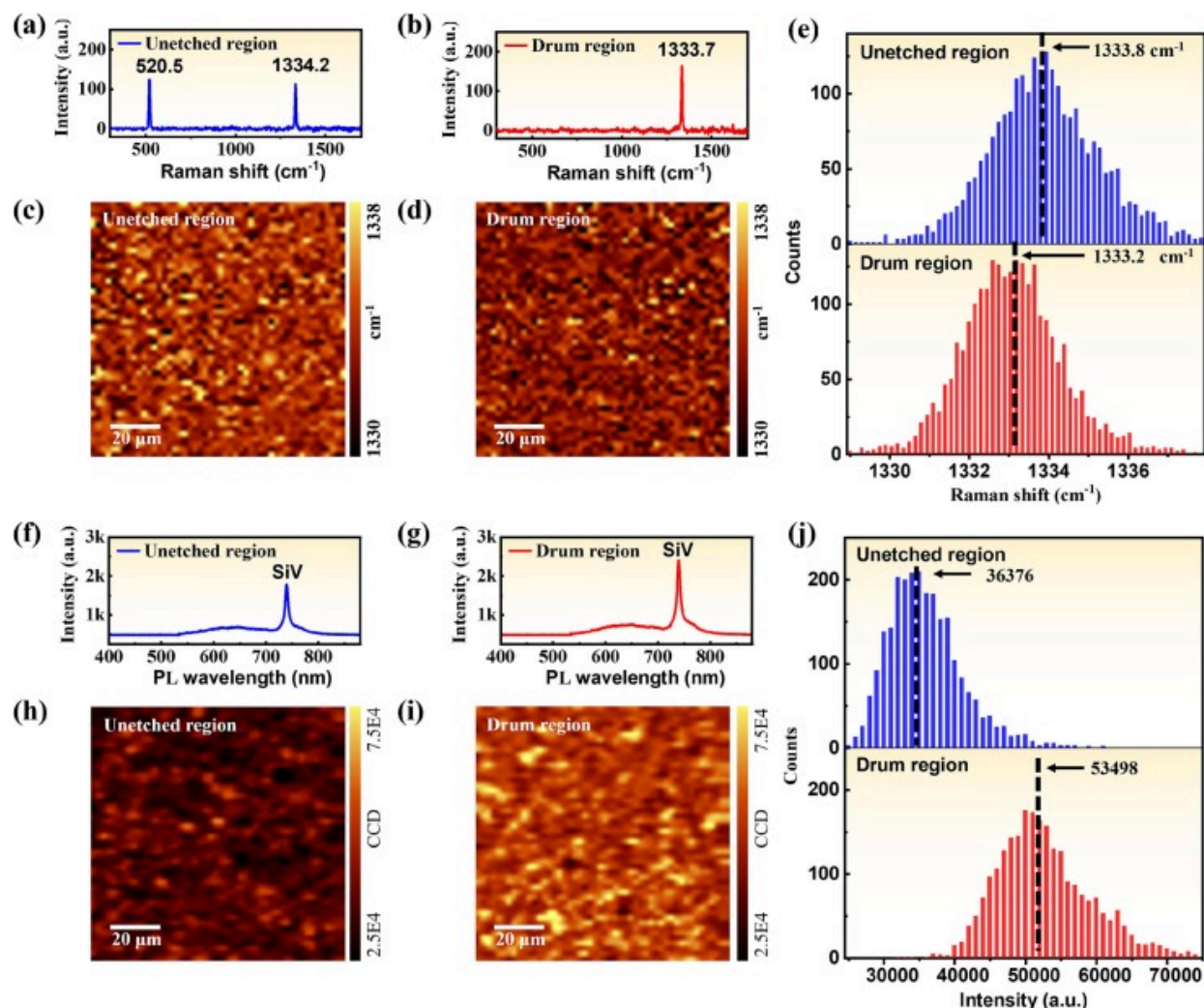
Fabricating diamond drums requires selective backside etching of Si substrate, which usually involves photolithography to create an etching mask, followed by wet chemical etching or deep silicon etching. In this study, we utilize a maskless technique with an ultrafast UV laser to engrave circular deep holes in the Si substrate, complemented by KOH etching to finalize the pattern. The laser pulse energy, laser repetition rate, and the number of processing passes were optimized to obtain desired structure. Fig. 3d shows a schematic of laser engraving operation, which created four circular holes of ~ 400 , 600, 800, and 1000 μm in diameters (the insets). The hole depths were determined to be $\sim 400\mu\text{m}$ in depth by vertically moving the focus planes of the confocal microscope. UV laser was selected in the study due to its short wavelength, minimal heat-affected zone, and high optical absorption. UV laser irradiation excites electrons to excited states and create a large number of photocarriers. The relaxation of photocarriers results in energy transfer to the lattices through electron-phonon coupling, leading to vigorous lattice vibration. This process is accompanied by plasma generation, and gives rise to powerful shock waves, thereby ejecting melted materials from the surface.

Fig. 3e(i) shows the statistical analysis of center position of the Raman characteristic peaks for Si in the laser engraved area (16,384 points on a $50\times 50\mu\text{m}^2$ region). We found that the Raman peak redshifts greatly from the standard position at 520cm^{-1} after laser ablation, indicating the formation of amorphous or nanocrystalline Si with very high tensile stress [40]. This observation was also confirmed by the PL spectra (Fig. 3e(ii)). The PL spectra collected by excitation with a 532nm laser reveals a largely broaden emission in the visible and near-infrared regions for the laser ablated areas, while the pristine area shows very low PL emission.

The laser ablated areas would serve as preferential locations for Si wet etching due to presence of irradiation induced defects and stresses [41]. Fig. 3f shows a schematic for the transformation of a circular hole into a trapezoidal pit on Si (100) surface due to

anisotropic etching [42]. KOH is a well-known etchant that preferentially attacks the Si (100) plane [43] and produce trapezoidal etch pits with sidewalls form a 54.7° with the (100) surface [44,45]. After etching, square-shaped diamond drums were eventually formed, regardless of the laser engraved circular holes. As a result, the length of the diamond drum is determined by the diameter and thickness of the initial circular holes. Experimentally, the sidewall lengths of the four drum resonators are determined to be 94, 296, 493, and 711 μm , respectively. We then measured the thickness of the diamond drums by white light optical interference [46]. Fig. 3g shows a typical optical reflection spectrum obtained on a diamond drum. The interference peaks and valleys occur at even and odd multiples of half-wavelengths. The thickness of the drums is determined to be 3.1 μm , in consistent with the thickness of the as-grown film measured by spiral micrometer.

Fig. 4 shows the Raman and the PL spectra of the diamond film on the drum and the unetched regions. Fig. 4a reveals an evident Raman peak of Si substrate at 520.5cm^{-1} for the unetched region. After etching, the 520.5cm^{-1} peak on the drum region disappears (Fig. 4b), indicating complete removal of the Si substrate. The diamond Raman peak at the unetched region locates at 1334.2cm^{-1} , which deviates slightly from that of diamond single crystal (1332.5cm^{-1}), indicating the existence of compressive stresses. In the drum region, the diamond Raman peak redshifts to 1333.7cm^{-1} . To ensure the reliability of the result, we acquired 2D Raman mapping images on the drum and the unetched regions, as shown in Fig. 4c and d. Fig. 4e shows an average Raman shift of 0.6cm^{-1} . The released stress is calculated to be $\sim 208\text{MPa}$ by using the empirical equation from the literature [47], $\sigma = \Delta\nu / 2.88\text{GPa}$, where σ is the residual stress, and $\Delta\nu$ is the Raman shift, respectively. The residual stress level depends on seeding density and CH_4 concentration during MPCVD growth, both of which can be further optimized for diamond film growth [48].



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Fig. 4. Typical Raman spectra and 2D Raman mapping images collected on the unetched (a, c) and drum (b, d) regions, respectively. (e) Statistical diagrams of the Raman peak positions of the two regions. Typical PL emission spectra and 2D PL mapping images collected on the unetched (f, h) and drum (g, f) regions, respectively. (j) Statistical diagrams of SiV PL emission intensity of two regions.

We also measured the PL spectra of the drum and the unetched regions. Fig. 4f reveals a PL emission peak at 739nm in the unetched region, which can be attributed to PL emission of silicon-vacancy (SiV) centers. This observation suggests diffusive Si incorporation into the diamond film during MPCVD growth. After etching, the PL emission intensity is greatly enhanced (Fig. 4g), as confirmed by 2D PL mapping at 739nm for the two regions (Fig. 4h and i). Fig. 4j shows the statistical distribution histogram of the PL emissions. We observe an average 47% increase of the PL intensity after Si removal. We tentatively attribute the reasons for this observation to the following factors: 1) reduced PL absorption by the Si substrate, 2) the formation of a Fabry-Perot cavity by the diamond drum for PL enhancement [49], and 3) possible activation of more doped Si atoms due to the high temperature induced by laser ablation [50].

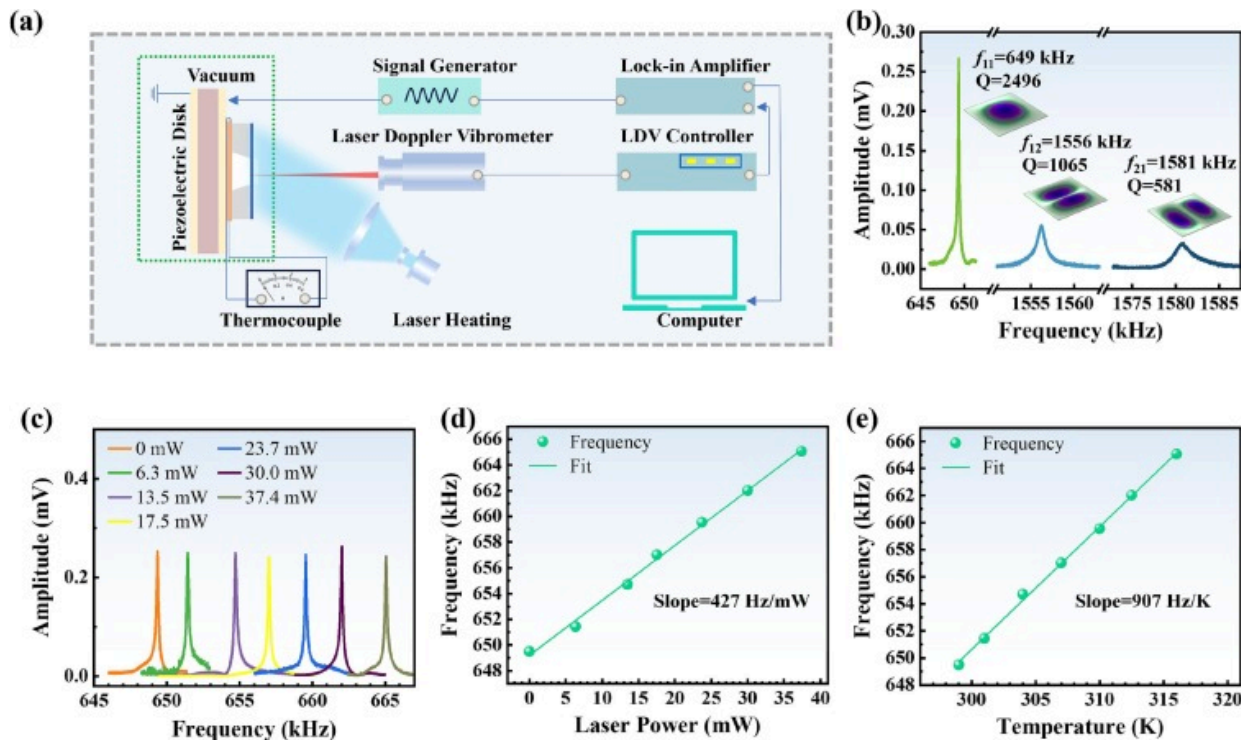
3.3. Temperature measurement performance of the diamond drum resonators

We characterized the temperature performance of the diamond drum resonator with a sidewall length of 296 μ m. We first measured the frequency-amplitude responses of the diamond drum resonator. Fig. 5a shows the experimental setup used to evaluate the vibrational behaviors of the resonator under vacuum condition (8 mTorr). The amplitude response curves are Lorentzian-fitted to extract resonant frequencies. Fig. 5b shows that the fundamental vibrational mode is located at 649kHz, characterized by a high quality (Q) factor of 2496. The second vibrational mode splits into two peaks located at 1.556 and 1.581MHz, most likely due to degeneracy lifting caused by slight variations of the side length. The two peaks exhibit reduced Q factors of 1065 and 581, due to formation of additional channels for energy dissipation by intermodal interactions [51]. Our calculation by using finite element method (FEM) shows a side length difference of, e.g. 1 μ m, could result in a peak splitting of 9kHz, in excellent agreement with the experimental result. Notably, we find that the frequency ratio of the fundamental mode to the second mode also deviates from the theoretical value 1:2 due possibly to the existence of residual stress in the diamond film [52,53]. The Young's modulus E of the

diamond film is estimated to be ~ 812 GPa through the following the formula [54]:

$$f_{m,n} = \sqrt{\frac{Eh^2 k_{m,n}^4}{48\pi^2 a^4 \rho(1-\nu^2)} + \frac{NJ_{m,n}}{4\rho ha^2}} \quad (1)$$

where $f_{m,n}$ is the natural frequency of the (m, n) modes, $k_{m,n}^2$ and $J_{m,n}$ is coefficient dependent on mode number ($k_{1,1}^2 = 35.99, k_{1,2}^2 = 73.41, J_{1,1} = 2.496, J_{1,2} = 5.906$), h is the thickness, a is the side length of the drum, ρ is the diamond density, ν is the Poisson ratio, and N is the in-plane loads per unit length of edge on the drum, respectively. The Young's modulus in this work is smaller than that of single crystal diamond because of relatively low microwave power and high methane concentration used during the MPCVD growth [55,56]. We also measured the first-order resonant frequencies of three drum resonators with side lengths of 234, 265, and 335 μm , and find that the results agree well with the thermotical values evaluated by using Eq. 1.



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Fig. 5. (a) Schematic diagram of the resonant frequency measurement setup; (b) Frequency responses of the first two modes of the diamond drum resonator (296 μm in length); (c) Frequency shift of the fundamental mode under laser heating of different powers; (d) Plot of the frequency shift of the fundamental mode versus laser power; (e) Plot of the frequency shift of the fundamental mode versus temperature.

We then measured the temperature dependent responses of the fundamental vibrational mode by heating the drum region with a 450nm continuous laser, equipped with a thermocouple to calibrate the temperature change ([Fig. 5a](#)). [Fig. 5c](#) reveals that increasing the laser power from 0 to 37.4mW results in a shift of the resonant frequency from 649.5 to 665.1 kHz. [Fig. 5d](#) and [e](#) are shows the frequency shift of the fundamental mode versus laser power and temperature. Obviously, the resonant frequency increases linearly with laser power and temperature. A linear fit of the fundamental frequency with respect to laser power and temperature was conducted by employing the least squares method. The results show a power sensitivity of 427Hz/mW and a temperature sensitivity of 907Hz/K. The temperature coefficient of frequency (TCF) is calculated to be 1396ppm/K that exceeds existing results in the literature [[57](#)], e.g. by using MoS₂ drum resonator [[58](#)] and Si₃N₄ double-ended-tuning-fork resonator [[59](#)]. The enhanced sensitivity could originate from various factors including material properties [[60](#)], residual stresses [[61,62](#)], and in particular the difference between the thermal expansion coefficients of the substrate and the thin film [[58,63](#)].

4. Conclusion

In summary, we report MPCVD growth of uniform diamond thin film on Si wafer through high-density NDs seeding via electrostatic adsorption of positively charged Ce-NDs colloids. The film exhibits a low RMS roughness of 70nm, a grain size of 1 μm , and a uniform stress distribution. Diamond drum resonators were fabricated by a combined utilization of backside engraving of Si wafer with ultrafast nanosecond 355nm laser and

anisotropic Si etching with KOH solution. It is found that removal of the underlying Si layer reduces the residual stresses by 208MPa and increases the SiV PL emission by 47%. The diamond drum resonator exhibits a high Q factor of 2496 and a temperature sensitivity of 907Hz/K (1396ppm/K). We expect the results report here could shed light in the design and fabrication of diamond-based MEMS sensors and devices.

CRediT authorship contribution statement

Yunchao Du: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Zhixin Zhao:** Methodology. **Haoran Li:** Methodology. **Biao Yan:** Methodology. **Duo Liu:** Writing – review & editing, Writing – original draft, Validation, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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